INTERNSHIP PROJECT REPORT

Learn To Build a Real-Time Job Analytics Portal

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1. Introduction

In today's digital era, data plays a very important role in decision-making across various industries. However, raw data by itself is not useful unless it is properly analyzed and presented in a clear and understandable format. This is where data analytics and visualization tools like Tableau become essential. They help in converting complex data into meaningful insights that can be easily interpreted.

This internship project, titled "Learn To Build a Real Time Job Analytics Portal," focuses on analyzing job-related data and presenting it through interactive visualizations. The main objective of this project was to gain hands-on experience in working with real-world datasets and to understand how different conditions and filters can be applied to extract relevant information.

During the internship, multiple tasks were performed using Tableau, each designed to test analytical thinking and problem-solving skills. These tasks involved creating different types of visualizations such as bar charts, scatter plots, tree maps, maps, and box-and-whisker plots. Each visualization required applying strict conditions, which helped in understanding how data behaves under different constraints.

Overall, this project helped in developing practical knowledge of data visualization, improving logical thinking, and gaining confidence in using Tableau for real-time data analysis. It also provided a clear understanding of how data can be transformed into valuable insights for better decision-making.

2. Background

Data visualization has become an essential part of modern data analysis, as it helps in understanding large datasets in a simple and effective way. Instead of going through raw data manually, visual representations such as charts and graphs make it easier to identify patterns, trends, and insights.

Tableau is a powerful data visualization tool that allows users to create interactive dashboards and reports without requiring advanced programming skills. It provides various features like filters, calculated fields, and different chart types, which make data analysis more flexible and efficient.

In this project, the dataset used contained job-related information such as job titles, company size, salary details, experience levels, work types, and locations. Working with this dataset required applying multiple filters and conditions to match real-world scenarios. These conditions helped in understanding how to manage and analyze structured data effectively.

The background of this internship mainly focused on applying theoretical knowledge of data analytics into practical use. By working on different tasks, it became possible to learn how to clean, filter, and visualize data in a meaningful way. This provided a strong foundation for building dashboards and extracting useful insights from data.

3. Learning Objectives

- To gain hands-on experience in analyzing real-world job-related data using Tableau and understand how raw data can be transformed into meaningful visual insights
- To learn how to handle datasets with multiple attributes by applying different filters and conditions to extract only the required information for analysis
- To develop the ability to create various types of visualizations such as bar charts, scatter plots, tree maps, maps, and box plots, and understand when each visualization is appropriate
- To improve logical thinking by working on tasks that involve strict and complex conditions, requiring careful implementation of multiple constraints simultaneously
- To understand the use of calculated fields in Tableau and how they help in applying custom logic, especially when dealing with string conditions, numerical limits, and time-based filters
- To build confidence in designing and organizing dashboards in a clean and structured manner, ensuring that the visualizations are easy to understand and interact with
- To enhance problem-solving skills by identifying errors in filters or calculations and correcting them to achieve the expected output
- To develop the ability to interpret data correctly and present insights in a clear, structured, and professional way

4. Activities and Tasks

During the internship, different analytical tasks were performed using Tableau by applying multiple conditions and filters to the dataset. Each task required careful implementation of given constraints to generate the expected visualization.

- **Task 1 – Preference vs Work Type (Bar Chart)**. Draw a bar chart between Preference and Work Type, where the Work Type must be "Intern", the Latitude must be below 10, and the Country name must not start with A, B, C, or D. The Job Title must be a single word with fewer than 10 characters, and the Company Size must be below 50,000. Only include postings where the Salary is above $9,000, the Experience is an even number, and the Job Posting Date is in an odd-numbered month. Sort the bars in descending order by count. The chart must only be displayed between 3 PM and 5 PM IST

- **Task 2 – Company Size vs Company Name (Scatter Plot)**. Draw a scatter plot between Company Size and Company Name, where Company Size < 50,000 and the Job Title must be exactly "Mechanical Engineer." The Experience must be > 5 years, the Country must be in Asia but not starting with "I," and the Salary must be above $50,000. Only include Work Type = Full-Time or Part-Time and Preference = Male. Additionally, filter only postings from the Job Portal = Idealist. Add one more condition: the Company Name must have at least 2 vowels. The chart must only be displayed between 3 PM and 5 PM IST

- **Task 3 – Top 10 Companies (Tree Map)**. Draw a tree map of the Top 10 Companies that have Role = Data Engineer and Job Title = Data Scientist. Exclude Asian countries and also exclude any countries starting with the letter "C." The Latitude must be below 10, and the Preference must be Female. The Qualification must be exactly B.Tech, and the Job Posting Date must be between 01/01/2023 and 06/01/2023. Only include postings from the Job Portal = LinkedIn, and keep only those where Company Size ≥ 10,000. Additional rule: the Contact Person name must end with a vowel. This chart must only be displayed between 3 PM and 5 PM IST

- **Task 4 – Qualification Drilldown Map (Map with Click).** Draw a map visualization where Qualification is B.Tech, M.Tech, or PhD and Work Type = Full-Time. The Country must be in Africa, and other countries should be ignored. The Job Title must

start with “D,” the Preference must be Male, and the Company Size must be greater than 80,000. The Contact Person must start with “A,” and the Job Portal must be Indeed. Use Latitude and Longitude to plot the jobs, and configure the chart so that clicking on a point opens a drilldown map showing the exact location. Extra condition: only include postings where the Salary Range > $20,000. The chart must only be displayed between 3 PM and 6 PM IST

- **Task 5 – India vs Germany Comparison (Stacked Bar)**. Draw a stacked bar chart comparing postings between India and Germany. Only include Qualification = B.Tech, Work Type = Full-Time, Experience > 2 years, and Job Title = Data Scientist, Art Teacher, or Aerospace Engineer. The Salary must be above $10,000, the Job Portal must be Indeed, and the Preference must be Female. Only include postings before 08/01/2023. Display India in Orange and Germany in Green. Additional rule: the Location field must not be blank, and the Company Name must have more than 8 characters. The chart must only be displayed between 3 PM and 5 PM IST

- **Task 6 – Work Type Salary Distribution (Box Plot)**. Draw a box-and-whisker plot between Work Type and Salary Range, where Work Type must be “Intern”, the Latitude is below 10, and the Country name must not start with A, B, C, or D. The Job Title must have fewer than 10 characters and be a single word. The Company Size must be < 50,000, and the Salary must be > $8,000. Additionally, only include postings where Experience is an even number, the Job Posting Date is between 2021–2023, and the Contact Person name must contain at least one “e.” The chart must only be displayed between 3 PM and 5 PM IST

5. Skills and Competencies

- During the internship, several technical and analytical skills were developed through hands-on experience with Tableau and real-world data.
- Gained practical understanding of how to create different types of visualizations such as bar charts, scatter plots, maps, tree maps, and box plots
- Improved ability to apply multiple filters and conditions accurately to meet specific task requirements
- Learned how to create and use calculated fields to implement custom logic and handle complex conditions
- Developed strong problem-solving skills by identifying errors in filters, calculations, and chart configurations and correcting them
- Enhanced analytical thinking by working with strict constraints and understanding how data behaves under different conditions
- Gained experience in organizing and structuring data for better visualization and interpretation
- Improved dashboard design skills by arranging charts and filters in a clear and user-friendly layout
- Built confidence in using Tableau as a tool for real-time data analysis and visualization

6. Feedback and Evidence

During the internship, regular feedback was received while working on different Tableau tasks. This feedback mainly focused on improving the accuracy of filters, correcting calculated field logic, and ensuring that each visualization matched the given conditions.

At the beginning, applying multiple conditions together was a bit confusing, especially when dealing with string-based filters and time-based conditions. With feedback and practice, it became easier to understand how to apply filters correctly and verify the output step by step.

The feedback also helped in identifying small mistakes such as incorrect filter types, wrong aggregation, or missing conditions. By correcting these errors, the overall understanding of Tableau improved, and the quality of the visualizations became better.

Evidence

The work completed during the internship is supported by the visualizations created in Tableau. Each task was implemented based on the given requirements, and the outputs demonstrate the correct application of filters, calculated fields, and chart configurations.

Screenshots of all tasks, including individual charts are included in the report as evidence. These screenshots clearly show how each condition was applied and how the final output was generated.

In some tasks, the output appears blank due to strict filtering conditions. This still serves as valid evidence, as it indicates that all conditions were applied correctly but no matching data was available in the dataset.

Each task's output has been documented separately to clearly demonstrate the implementation and results.

Task 1 – Preference vs Work Type (Bar Chart)

In this task, a bar chart was created using Preference as the dimension and the count of job postings as the measure. The Work Type was restricted to Intern, and all the given conditions such as latitude below 10, exclusion of countries starting with A, B, C, or D, job title constraints, company size limit, salary threshold, and even experience were applied.

The chart was sorted in descending order and displayed based on the specified time condition. The final output was generated after applying all filters correctly.

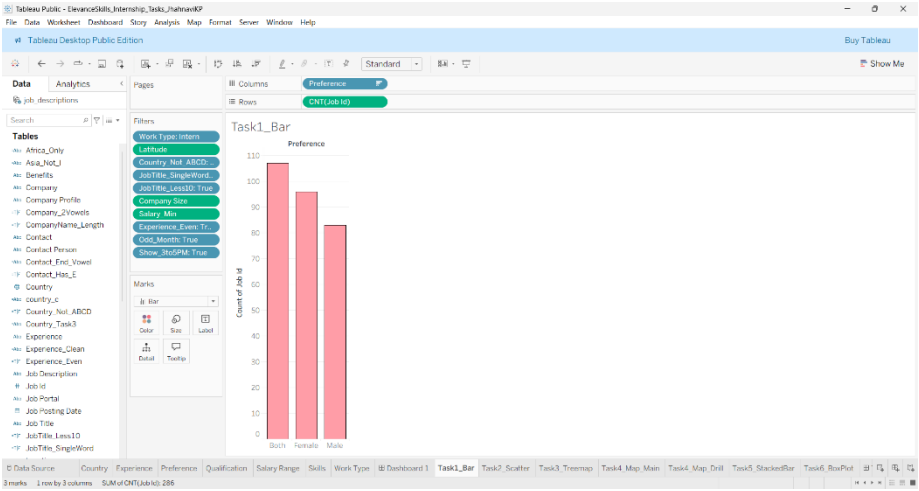

Fig 1: Task 1 – Preference vs Work Type (Bar Chart)

Task 2 – Company Size vs Company Name (Scatter Plot)

In this task, a scatter plot was created to analyze the relationship between company size and company name. Company Name was used as the dimension, and Company Size was used as the measure. All the required conditions such as job title being Mechanical Engineer, experience greater than 5 years, salary above $50,000, and company size below 50,000 were applied. Additional filters like country restricted to Asia excluding those starting with "I", work type limited to Full-Time and Part-Time, preference set to Male, and job portal as Idealist were implemented. A condition to ensure company names contain at least two vowels was also applied, along with the time-based filter.

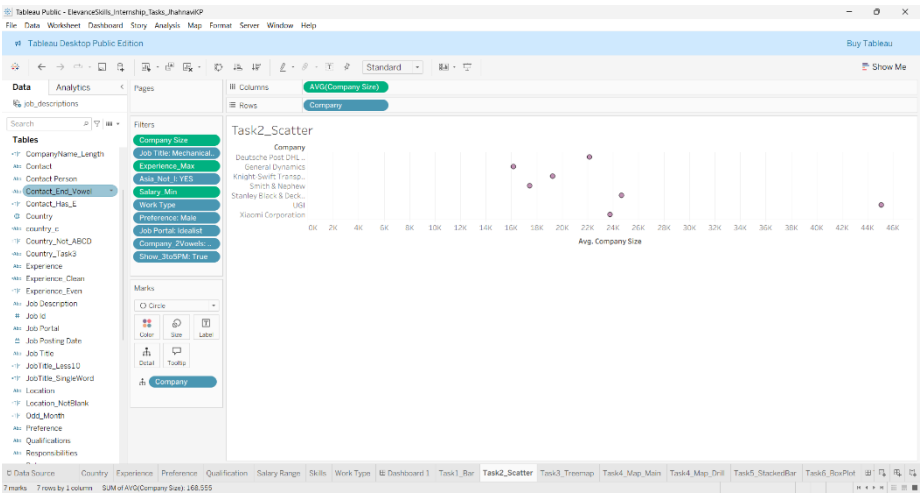

Fig 2: Task 2 – Company Size vs Company Name (Scatter Plot)

Task 3 – Top 10 Companies (Tree Map)

In this task, a tree map was created to display the top 10 companies based on job postings. The visualization was built using Company as the dimension and count of job postings as the measure. Conditions such as role being Data Engineer and job title being Data Scientist were applied.

Additional filters included excluding Asian countries and countries starting with "C", latitude below 10, preference set to Female, qualification as B.Tech, job posting date within the given range, job portal as LinkedIn, and company size greater than or equal to 10,000. A condition was also applied to ensure the contact person name ends with a vowel.

After applying all the conditions, the output was not visible because no records in the dataset satisfied all the requirements simultaneously.

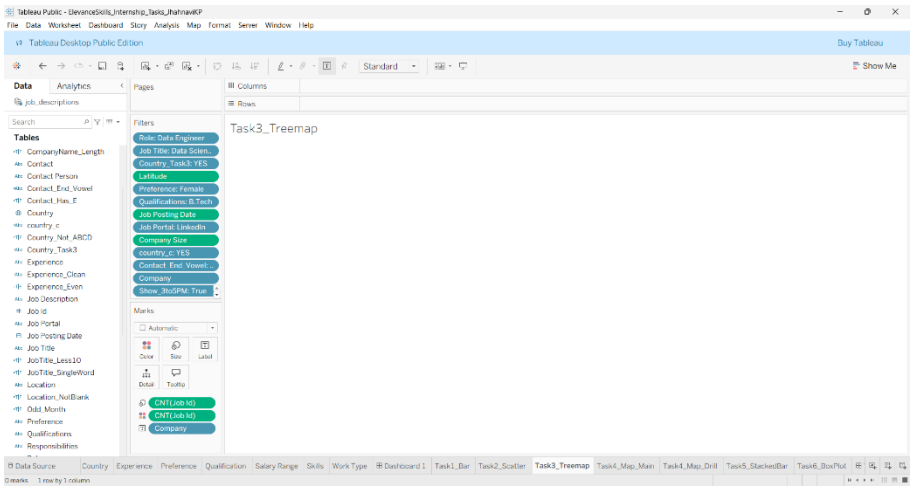

Fig 3: Task 3 – Top 10 Companies (Tree Map)

Task 4 – Qualification Drilldown Map (Map Visualization)

In this task, a map visualization was created using latitude and longitude to represent job locations. The visualization was built by applying conditions such as qualification being B.Tech, M.Tech, or PhD and work type restricted to Full-Time.

Additional filters included selecting countries only from Africa, job title starting with "D", preference set to Male, company size greater than 80,000, job portal as Indeed, and contact person name starting with "A". A salary condition greater than $20,000 was also applied. A drill-down feature was implemented to allow users to click on a location and view detailed information in another map.

After applying all the required conditions, no data was displayed due to the strict filtering criteria.

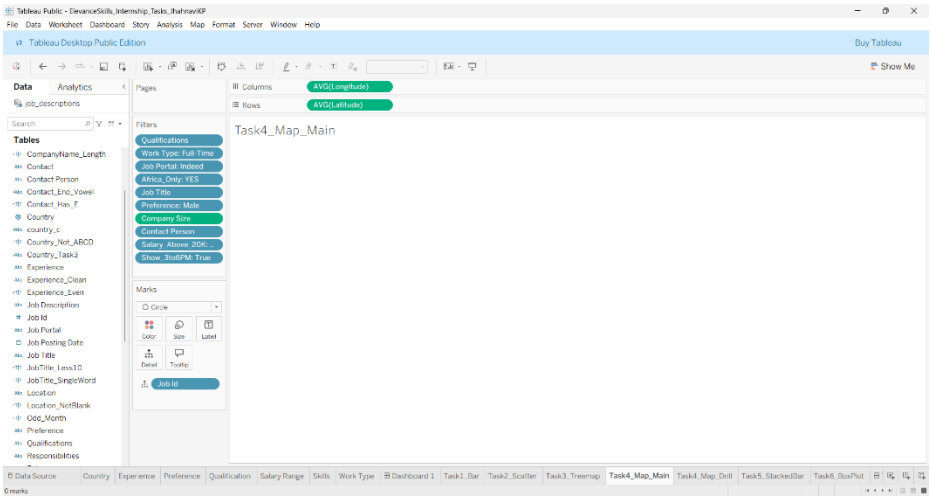

Fig 4.1: Task 4 – Drilldown Map_Main

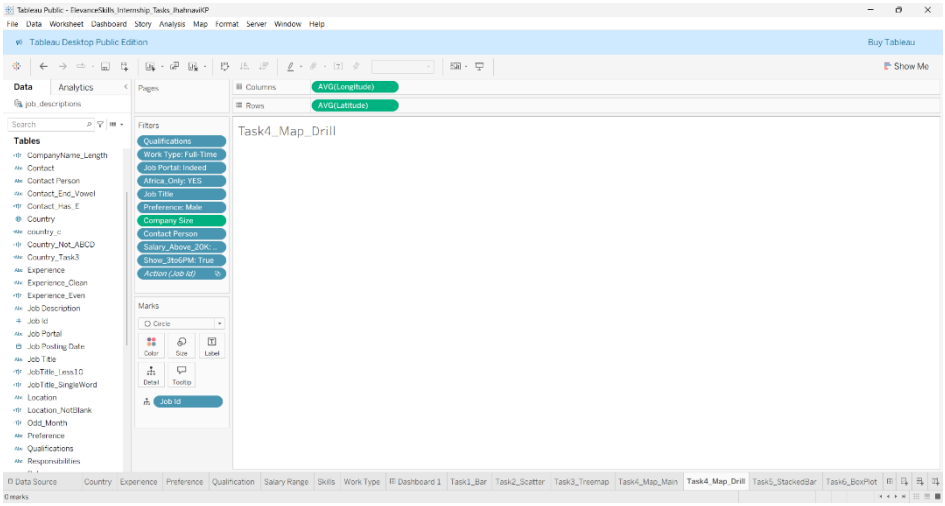

Fig 4.2: Task 4 – Drilldown Map_Drill

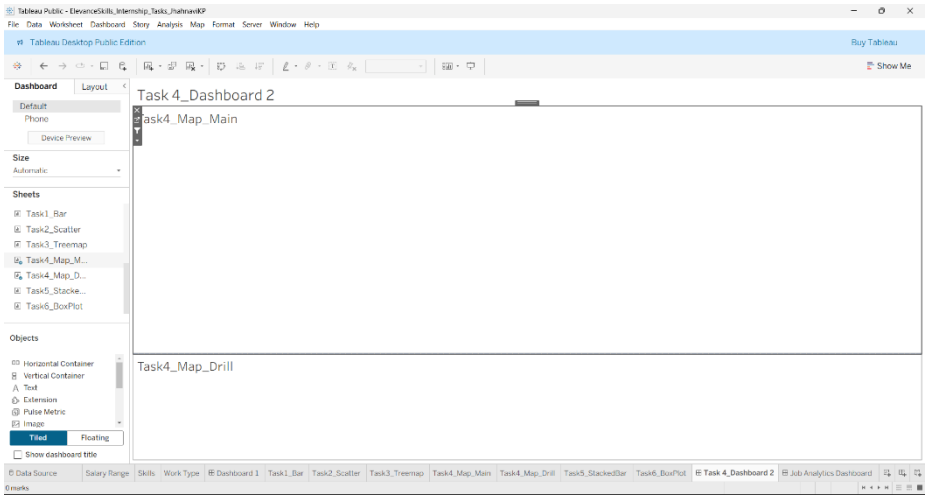

Fig 4.3: Task 4 – Drilldown Map_Dashboard

Task 5 – India vs Germany Comparison (Stacked Bar Chart)

In this task, a stacked bar chart was created to compare job postings between India and Germany. Country was used as the dimension, and the count of job postings was used as the measure. All required conditions such as qualification being B.Tech, work type as Full-Time, experience greater than 2 years, and job titles limited to Data Scientist, Art Teacher, and Aerospace Engineer were applied. Additional filters included salary above $10,000, job portal as Indeed, preference set to Female, job posting date before 08/01/2023, non-null location values, and company name length greater than 8 characters.The chart was customized by assigning different colors to each country, with India represented in orange and Germany in green.

Due to the strict filtering conditions, data for India was not available, and only Germany was visible in the final output. The visualization was displayed based on the specified time condition.

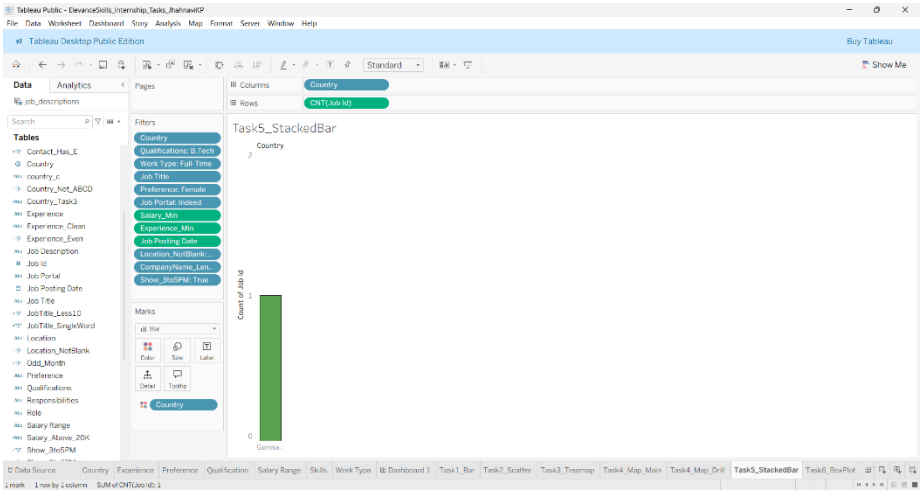

Fig 5: Task 5 – India vs Germany Comparison (Stacked Bar Chart)

Task 6 – Work Type Salary Distribution (Box Plot)

In this task, a box-and-whisker plot was created to analyze the salary distribution for Intern roles. Work Type was used as the dimension, and salary values were represented using a numeric field derived from the salary range.

All required conditions such as latitude below 10, exclusion of countries starting with A, B, C, or D, job title constraints, company size below 50,000, salary above $8,000, and even-numbered experience were applied. Additional filters included job posting date between 2021 and 2023 and contact person name containing at least one “e”.

After applying all conditions, the visualization showed a limited number of data points due to strict filtering, resulting in a minimal box plot. The chart was displayed based on the specified time condition.

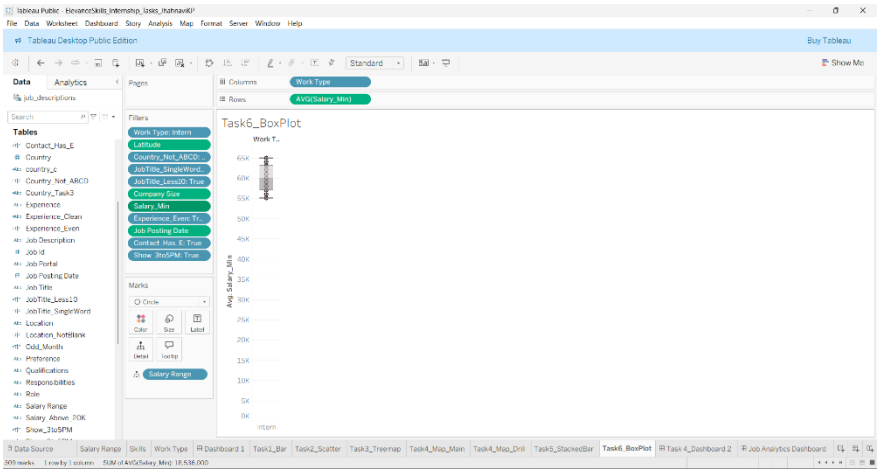

Fig 6: Task 6 – Work Type Salary Distribution (Box Plot)

Final Dashboard

The final dashboard combines all tasks into a single view for better presentation. Task 1, Task 5, and Task 6 were arranged vertically as they have similar structures, while Task 2 was placed horizontally for better visibility. Task 3 and Task 4 were included without changes, even though they do not show output, as all conditions were correctly applied and the dataset did not meet the requirements.

Interactive filters such as Work Type, Preference, Qualification, and Salary were added to improve usability. The layout is kept simple and organized for better understanding.

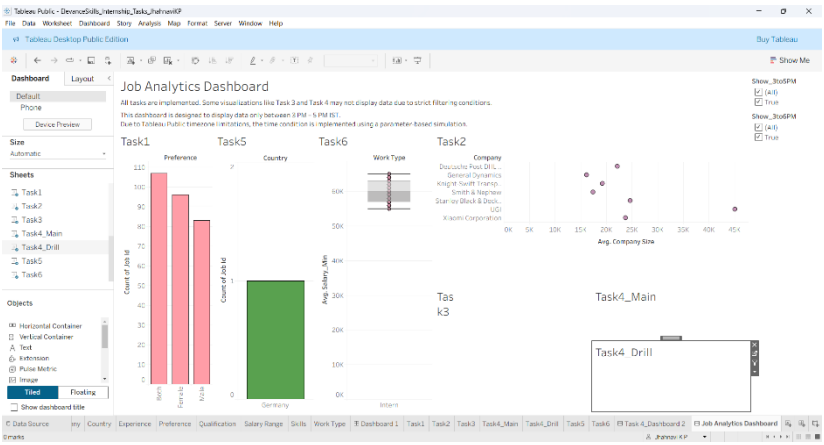

Fig 7: Final Dashboard

7. Challenges and Solutions

During the internship, several challenges were faced while working on different Tableau tasks. These challenges mainly occurred due to strict filtering conditions and handling multiple constraints at the same time.

- **Challenge:** Applying multiple filters together resulted in blank outputs for some tasks. **Solution:** All conditions were verified step by step, and it was confirmed that the absence of data was due to dataset limitations.
- **Challenge:** Difficulty in working with non-numeric salary range values for the box plot.

  Solution: A numeric field (Salary_Min) was used to represent salary values for proper visualization.
- **Challenge:** Errors in calculated fields due to incorrect logic or syntax. **Solution:** The formulas were corrected using appropriate Tableau functions and tested for accuracy.
- **Challenge:** Confusion in selecting the correct type of filters. **Solution:** Filters were applied carefully based on requirements and verified after each step.
- **Challenge:** Managing multiple conditions without missing any requirement. **Solution:** A step-by-step approach was followed, and each condition was cross-checked with the task.
- **Challenge:** Limited data availability after applying strict conditions. **Solution:** Verified that all filters were correct and understood that fewer data points affect the output.
- **Challenge:** Difficulty in arranging charts and filters in a proper dashboard layout. **Solution:** Adjusted the layout and spacing to create a clean and user-friendly dashboard.

8. Outcomes and Impact

This internship helped in gaining practical experience in working with real-world data using Tableau. By completing multiple tasks, it became easier to understand how to analyze data, apply filters, and create meaningful visualizations.

The project improved analytical thinking, as each task required careful application of conditions and attention to detail. Working with strict filtering requirements also helped in understanding how data availability affects the final output.

This experience also increased confidence in using Tableau for creating dashboards and handling different types of visualizations. It helped in developing problem-solving skills and a better approach to working with complex data conditions.

Overall, the internship had a positive impact on both technical and analytical skills, which will be useful for future projects and career opportunities in data analysis.

9. Conclusion

The internship provided valuable hands-on experience in data analysis and visualization using Tableau. Through the completion of various tasks, it became possible to understand how to work with real-world datasets, apply multiple conditions, and create different types of visualizations.

The project helped in improving logical thinking, problem-solving skills, and attention to detail while working with complex requirements. It also highlighted how data availability and filtering conditions can affect the final output of visualizations.

Overall, this internship was a useful learning experience that strengthened both technical and analytical skills. The knowledge gained will be helpful for future academic work and career opportunities in the field of data analytics.

Hosted Links

1. **Tableau Public**
   https://public.tableau.com/app/profile/jhahnavi.k.p/viz/ElevanceSkills_Internship_Tasks_JhahnaviKP/JobAnalyticsDashboard?publish=yes
2. **Netlify**
   https://jobbbanalyticsssportalll.netlify.app/